

## CSA05-DATABASE MANAGEMENT SYSTEMS

### LAB OBSERVATION

#### EXPERIMENT:06

#### AIM:

To view the records from the tables using SELECT commands with BETWEEN, IN, Aggregate functions.

#### QUESTION:

1. List the type of the courses “Statistics” and “Programming”

#### SQL QUERIES:

```
mysql> SELECT CourseName, CourseType
-> FROM COURSE
-> WHERE CourseName IN ('Statistics','Programming');
+-----+-----+
| CourseName | CourseType |
+-----+-----+
| Statistics | Theory     |
| Programming | Practical  |
+-----+-----+
2 rows in set (0.00 sec)
```

2. The instructor wants to know the CourseNos whose scores are in the range 50 to 80

#### SQL QUERIES:

```
mysql> SELECT CourseName, CourseType
-> FROM COURSE
-> WHERE CourseName IN ('Statistics','Programming');
+-----+-----+
| CourseName | CourseType |
+-----+-----+
| Statistics | Theory     |
| Programming | Practical  |
+-----+-----+
2 rows in set (0.00 sec)
```

3. Find the average mark of “C002”.

#### SQL QUERIES:

```
mysql> SELECT AVG(Score) AS Average_Mark
-> FROM SCORE
-> WHERE CourseNo = 'C002';
+-----+
| Average_Mark |
+-----+
| 67.20        |
+-----+
1 row in set (0.00 sec)
```

4. List the maximum, minimum mark for “C002”

#### SQL QUERIES:

```
mysql> SELECT MAX(Score) AS Max_Mark,
-> MIN(Score) AS Min_Mark
-> FROM SCORE
-> WHERE CourseNo = 'C002';
```

| Max_Mark | Min_Mark |
|----------|----------|
| 92       | 45       |

1 row in set (0.00 sec)

- List the maximum, minimum, average mark for each course

#### SQL QUERIES:

```
mysql> SELECT CourseNo, MAX(Score) AS Max_Mark,
-> MIN(Score) AS Min_Mark, AVG(Score) AS Avg_Mark
-> FROM SCORE
-> GROUP BY CourseNo;
```

| CourseNo | Max_Mark | Min_Mark | Avg_Mark |
|----------|----------|----------|----------|
| C001     | 95       | 55       | 72.50    |
| C002     | 92       | 45       | 67.20    |
| C003     | 88       | 60       | 74.00    |
| C004     | 90       | 50       | 65.60    |
| C005     | 85       | 40       | 63.25    |

5 rows in set (0.00 sec)

- List the name of the courses and average mark of each course.

#### SQL QUERIES:

```
mysql> SELECT c.CourseName, AVG(s.Score) AS Avg_Mark
-> FROM COURSE c
-> JOIN SCORE s ON c.CourseNo = s.CourseNo
-> GROUP BY c.CourseName;
```

| CourseName        | Avg_Mark |
|-------------------|----------|
| Statistics        | 72.50    |
| Programming       | 67.20    |
| Database          | 74.00    |
| Networks          | 65.60    |
| Operating Systems | 63.25    |

5 rows in set (0.00 sec)

- Calculate the sum of all the scores.

#### SQL QUERIES:

```
mysql> SELECT SUM(Score) AS Total_Score
-> FROM SCORE;
```

| Total_Score |
|-------------|
| 1689        |

1 row in set (0.00 sec)

- How many students are registered for each course? Display the course description and the number of students registered in each course.

#### SQL QUERIES:

```
mysql> SELECT c.CourseName, COUNT(*) AS Number_Of_Students
-> FROM COURSE c
-> JOIN REGISTRATION r ON c.CourseNo = r.CourseNo
-> GROUP BY c.CourseName;
```

| CourseName        | Number_Of_Students |
|-------------------|--------------------|
| Statistics        | 4                  |
| Programming       | 5                  |
| Database          | 4                  |
| Networks          | 3                  |
| Operating Systems | 4                  |

```
5 rows in set (0.00 sec)
```

9. How many courses did each student register for?

#### SQL QUERIES:

```
mysql> SELECT s.RegNo, s.SName, COUNT(r.CourseNo) AS Number_Of_Courses
-> FROM STUDENT2 s
-> JOIN REGISTRATION r ON s.RegNo = r.RegNo
-> GROUP BY s.RegNo, s.SName;
```

| RegNo     | SName   | Number_Of_Courses |
|-----------|---------|-------------------|
| 191711341 | Aniha   | 3                 |
| 191711341 | Bharath | 2                 |
| 191711343 | Charan  | 3                 |
| 191711344 | Divya   | 2                 |
| 191711345 | Eshwar  | 3                 |

```
5 rows in set (0.00 sec)
```

#### RESULT:

The records from the tables are displayed using SELECT commands with WHERE Clause and Pattern matching.